K L UNIVERSITY

FRESHMAN ENGINEERING DEPARTMENT

A Project Based Lab Report

On

SMART-HEALTH-MEDICAL-HISTORY-TRACKER-WELLNESS-ANALYTICS-SYSTEM

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CERTIFICATE

This is to certify that the project based laboratory report entitled "SMART HEALTH & MEDICAL HISTORY TRACKER + WELLNESS ANALYTICS SYSTEM (SHM-WAS)" submitted by **G.YashwanthKumar** bearing Regd. No. **2500030779** to the **Department of COMPUTER SICENCE ENGINEERING , KL University** in partial fulfillment of the requirements for the completion of a project based Laboratory in "Python Programming (25SC2007)" course in I B Tech I Semester, is a Bonafede record of the work carried out by him/her under my supervision during the academic year 2025 – 2026.

PROJECT SUPERVISOR

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ABSTRACT

This project is aimed at developing a Smart Health & Medical History Tracker along with a Wellness Analytics System (SHM-WAS) for users. The system focuses on providing a digital and centralized platform to store, manage, and analyze personal health data efficiently. In today's fast-paced world, maintaining physical medical records can be difficult and unreliable; therefore, this system ensures easy accessibility, better organization, and secure handling of health information. It integrates modern technology to help users keep track of their medical history and overall wellness in a smarter and more convenient way.

The Smart Health system helps users maintain records such as past medical reports, prescriptions, allergies, and ongoing treatments. It also provides features for tracking daily health parameters like heart rate, steps, sleep patterns, and other wellness indicators. By analyzing this data, the system offers useful insights and suggestions to improve overall health and lifestyle.

The project's objective is to create an easy-to-use and reliable platform that supports both patients and healthcare providers. Users can update their health data regularly, receive reminders for medications or check-ups, and monitor their wellness trends over time. The system also ensures data privacy and security.

Overall, SHM-WAS helps in improving health awareness, enables better decision-making, and promotes a healthier lifestyle through smart tracking and analytics.

Key Words: health tracking, medical history, wellness analytics, digital healthcare, smart system.

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INTRODUCTION

In recent years, the importance of maintaining accurate and accessible health records has increased significantly due to the growing awareness of personal healthcare and wellness. Traditional methods of storing medical records in physical form are often inefficient, difficult to manage, and prone to loss or damage. With the advancement of digital technologies, there is a need for a smart and reliable system that can store, organize, and analyze health-related data effectively.The Smart Health & Medical History Tracker with Wellness Analytics System (SHM-WAS) is designed to address these challenges by providing a centralized digital platform for managing personal health information. This system enables users to securely store their medical records, including prescriptions, test reports, allergies, and treatment history, in an organized manner. It eliminates the need for carrying physical documents and ensures that important health information is available whenever required.

- · Importance of digital healthcare systems in modern life
- · Problems with traditional paper-based medical records
- · Need for easy access to health data anytime and anywhere
- · Use of technology for better health monitoring
- · Role of data analytics in improving lifestyle and health decisions
- · Benefits of maintaining complete medical history in one place
- · Support for both patients and doctors

AIM

The aim of this project is to develop a Smart Health & Medical History Tracker with a Wellness Analytics System (SHM-WAS) that enables users to store, manage, and access their medical records digitally while tracking daily health parameters and providing useful insights to improve overall health and lifestyle.

- · To develop a digital system for storing and managing medical records
- · To provide easy access to health data anytime and anywhere
- · To track daily health parameters like heart rate, steps, and sleep
- · To analyze health data and provide useful insights
- · To improve overall health awareness among users
- · To reduce dependency on physical medical records
- · To provide reminders for medications and regular check-ups
- · To ensure data privacy and security of user information

ADVANTAGES

- · Easy storage of medical records in digital form
- · Quick access to health information anytime and anywhere
- · Reduces the risk of losing important medical documents
- · Helps in tracking daily health activities and wellness
- · Provides useful insights to improve health and lifestyle
- · Saves time during doctor visits by having complete records ready
- · Sends reminders for medicines and regular check-ups
- · Improves health awareness among users

DISADVANTAGES

- · Requires internet connection for full functionality
- · Risk of data breaches if security is not properly maintained
- · Dependence on digital devices like smartphones or computers
- · Initial setup and data entry may take time
- · May be difficult to use for non-technical or elderly users
- · Accuracy depends on correct data input by the user
- · Maintenance and updates are required regularly
- · Possible technical issues like app crashes or bugs
- · Privacy concerns related to storing sensitive health data online

FUTURE IMPLEMENTATION

In the future, the Smart Health & Medical History Tracker with Wellness Analytics System (SHM-WAS) can be enhanced by integrating advanced technologies such as artificial intelligence and machine learning to provide more accurate health predictions and personalized recommendations. The system can be connected with wearable devices like smartwatches and fitness bands for real-time health monitoring. It can also include features such as online doctor consultation, emergency alert systems, and integration with hospital databases for better coordination. Additionally, improvements in data security using advanced encryption techniques can be implemented to ensure higher privacy protection.

SYSTEM REQUIREMENTS

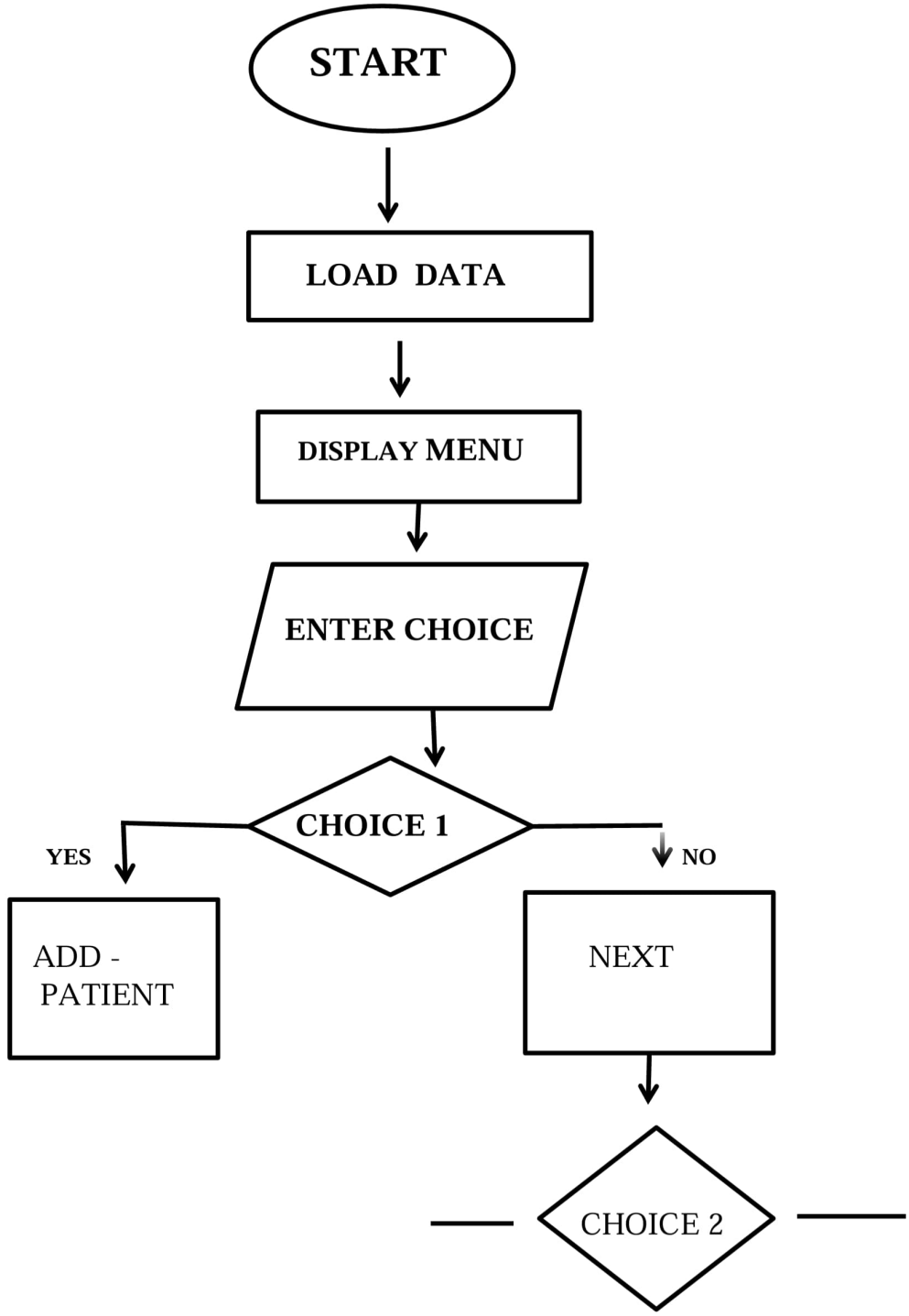
➢ SOFTWARE REQUIREMENTS:

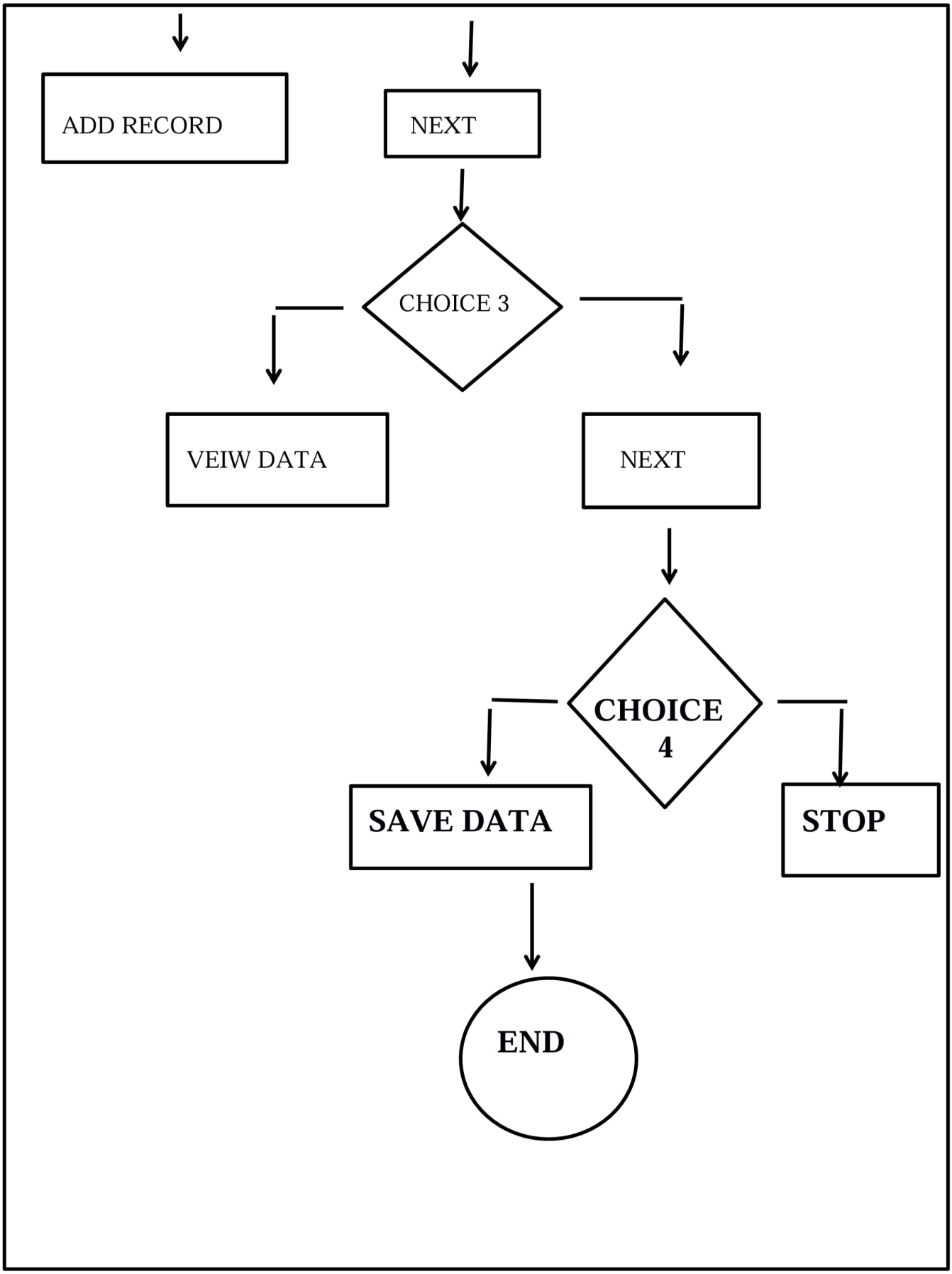
The system requires programming languages such as Python, Java, or JavaScript for development. Front-end technologies like HTML, CSS, and JavaScript are used for designing the user interface, while backend frameworks such as Node.js or Django handle server-side operations. A database like MySQL or MongoDB is used to store health data. Development tools such as Visual Studio Code or Android Studio and web browsers like Google Chrome are used for coding and testing.

➢ HARDWARE REQUIREMENTS:

A computer or smartphone with internet connection is required. Basic devices like keyboard and mouse are used. Wearable devices can be added for health tracking.

DATA FLOW DIAGRAM




ADD RECORD
NEXT
CHOICE 3
VEIW DATA
NEXT
CHOICE 4
SAVE DATA
STOP
END

ALGORITH

Step-1: Start.

Step-2: Read user details (name, age, health information).

Step-3: Login or register user.

Step-4: Display menu.

1. Enter health data.
2. View medical history.
3. Analyze health data.
4. View reports.

Step-5: Read user selection.

Step-6: If user selects Enter Health Data.
1. Read health parameters (heart rate, steps, sleep, etc.).
2. Validate data.
3. Store data in database.

Step-7: If user selects View Medical History.
1. Retrieve data from database.
2. Display medical records.

Step-8: If user selects Analyze Data.
1. Process stored data.
2. Generate insights.

Step-9: If user selects View Reports.
1. Display health reports.

Step-10: Provide reminders and notifications.

Step-11: Thank you.

Step-12: Stop.

IMPLEMENTATION

```
import json

# File to store data
FILE_NAME = "health_data.json"

# Load existing data
def load_data():
    try:
        with open(FILE_NAME, "r") as file:
            return json.load(file)
    except:
        return {}

# Save data
def save_data(data):
    with open(FILE_NAME, "w") as file:
        json.dump(data, file, indent=4)

# Add new patient
def add_patient(data):
    patient_id = input("Enter Patient ID: ")
    name = input("Enter Name: ")
    age = input("Enter Age: ")

    data[patient_id] = {
        "name": name,
        "age": age,
        "records": []
    }

    print("✅Patient added successfully!")

# Add health record
def add_record(data):
    patient_id = input("Enter Patient ID: ")
```

```
 if patient_id not in data:
    print("❌Patient not found!")
    return

 bp = input("Enter Blood Pressure: ")
 sugar = input("Enter Sugar Level: ")
 weight = input("Enter Weight: ")

 record = {
    "bp": bp,
    "sugar": sugar,
    "weight": weight
 }

 data[patient_id]["records"].append(record)
 print("✅Record added successfully!")

 # View patient data
def view_patient(data):
 patient_id = input("Enter Patient ID: ")

 if patient_id not in data:
    print("❌Patient not found!")
    return

 print("\n--- Patient Details ---")
 print("Name:", data[patient_id]["name"])
 print("Age:", data[patient_id]["age"])

 print("\n--- Health Records ---")
 for record in data[patient_id]["records"]:
    print(record)

# Simple analytics
def analytics(data):
 patient_id = input("Enter Patient ID: ")

 if patient_id not in data:
    print("❌Patient not found!")
    return
```

```
    records = data[patient_id]["records"]

    if not records:
        print("No records available.")
        return

    total_bp = 0
    total_sugar = 0
    count = len(records)

    for r in records:
        total_bp += int(r["bp"])
        total_sugar += int(r["sugar"])

    print("\n--- Analytics ---")
    print("Average BP:", total_bp // count)
    print("Average Sugar:", total_sugar // count)

# Main menu
def main():
    data = load_data()

    while True:
        print("\n==== SHM-WAS MENU ====")
        print("1. Add Patient")
        print("2. Add Health Record")
        print("3. View Patient Data")
        print("4. Analytics")
        print("5. Exit")

        choice = input("Enter choice: ")

        if choice == "1":
            add_patient(data)
        elif choice == "2":
            add_record(data)
        elif choice == "3":
            view_patient(data)
        elif choice == "4":
```

```
        analytics(data)
    elif choice == "5":
        save_data(data)
        print("Data saved. Exiting...")
        break
    else:
        print("Invalid choice!")

# Run program
 main()
```

INTEGRATION AND SYSTEM TESTING OUTPUTS

Screen Shots:

```
Output                                                    Clear

==== SHM-WAS MENU ====
1. Add Patient
2. Add Health Record
3. View Patient Data
4. Analytics
5. Exit
Enter choice: 2
Enter Patient ID: 5
❌ Patient not found!

==== SHM-WAS MENU ====
1. Add Patient
2. Add Health Record
3. View Patient Data
4. Analytics
5. Exit
Enter choice: 1
Enter Patient ID: 23
Enter Name: yesh
Enter Age: 3
✅ Patient added successfully!
```

CONCLUSION

The *Smart Health & Medical History Tracker with Wellness Analytics System (SHM-WAS)* successfully demonstrates an efficient and reliable approach to managing personal health records digitally. This system helps users securely store, access, and monitor their medical history anytime and anywhere, reducing the dependency on physical documents.The integration of wellness analytics enables users to gain valuable insights into their health conditions, promoting better awareness and proactive healthcare decisions. The system ensures data accuracy, accessibility, and security, making it a practical solution for modern healthcare management.Overall, this project highlights the importance of digital transformation in the healthcare sector and provides a user-friendly platform that can be further enhanced with advanced features like real-time monitoring, AI-based predictions, and cloud integration for improved performance and scalability.